Tissue-Specific DEG Analysis

Instruction Manual v2 - Corrected Unnormalized DESeq2 Data

Generated by Biomni | *2026-07-10*

> **For Beginners**
>
> This manual explains how to analyze tissue-specific gene expression changes in Arabidopsis plants exposed to different types of radiation. We use 6 separate experiments from NASA GeneLab, covering gamma rays (Co-60, Cs-137), simulated cosmic rays (GCR), and a DNA repair mutant (sog1-1). Each section has callout boxes like this one for key concepts.

> **For Advanced Readers**
>
> This version uses raw unnormalized counts as DESeq2 input (corrected from the previous version which used normalized data). DEG threshold: padj < 0.05 AND |log2FC| >= 1. Tissue assignment via AtGenExpress atlas (Schmid et al. 2005) with Tau tissue-specificity index (threshold 0.6). Cross-comparison tests use Cochran-Mantel-Haenszel and chi-square frameworks.

1. Introduction: Arabidopsis Radiation Response

Plants grown in space are exposed to ionizing radiation that differs from Earth's background. NASA GeneLab has deposited multiple Arabidopsis RNA-seq studies examining how plants respond to different radiation types. This analysis covers 6 experimental comparisons from 5 OSD studies (OSD-498, 508, 510, 658, 782), encompassing 158 samples across 3 radiation sources, 4 dose levels, and 3 genotypes.

The 6 Experimental Comparisons

#	Label	Study	Contrast	DESeq2 Design
1	radiation_effect	OSD498+510	Gamma Co-60 100Gy vs none (WT)	~ Study + Genotype + Radiation
2	genotype_interaction	OSD508+510	sog1-1 x radiation interaction	~ Study + Genotype + Radiation + Geno:Rad
3	GCR40	OSD658	Simulated GCR 40cGy vs none (WT)	~ Radiation
4	GCR80	OSD658	Simulated GCR 80cGy vs none (WT)	~ Radiation
5	Cs137_100cGy	OSD782	Gamma Cs-137 100cGy vs none (WT)	~ Timepoint + Radiation
6	Cs137_10cGy	OSD782	Gamma Cs-137 10cGy vs none (WT)	~ Timepoint + Radiation

> **Key Concept: Why 6 Separate Comparisons?**
>
> Each study uses different radiation sources, doses, and genotypes. Pooling them into one model would average together biologically distinct exposures. Instead, we run focused comparisons within studies that share a consistent design, treating 'Study' as a batch covariate where needed.

Advanced Note: Study as a batch effect

Different sequencing runs, years, and library preparations mean 'Study' itself can explain a large chunk of gene expression variance. In pooled comparisons (1 and 2), Study is included as a blocking term in the DESeq2 design formula. Single-study comparisons (3-6) do not need this because there is no cross-study batch to account for.

2. DESeq2 and the Raw vs Normalized Counts Issue

DESeq2 is the standard tool for differential expression analysis of RNA-seq count data. It models read counts with a negative binomial distribution and performs its own internal normalization (median-of-ratios size factors). This means DESeq2 expects raw, unnormalized integer counts as input.

> **Methodological Lesson: Raw vs Normalized**
>
> The first version of this analysis used GeneLab's normalized counts table as DESeq2 input. This is methodologically incorrect because DESeq2's dispersion estimation and statistical testing assume raw count data. The corrected version uses raw unnormalized counts (RSEM/STAR output), which is the statistically proper input. The impact: 6,942 DEGs (old) vs 822 DEGs (new) for the same comparison, with 99.8% of lost DEGs having |log2FC| < 1 - they were called significant only because the old threshold had no fold-change cutoff.

DEG Threshold

We define differentially expressed genes (DEGs) as those with adjusted p-value (padj) < 0.05 AND absolute log2 fold change >= 1 (at least 2-fold change). This dual threshold ensures both statistical significance and biological meaningfulness - a gene can have a tiny p-value but a negligible fold change that is unlikely to matter biologically.

Advanced Note: Why |log2FC| >= 1?

The old analysis used padj < 0.05 with no fold-change cutoff, calling genes with |log2FC| as low as 0.057 as DEGs. While statistically significant, a 4% change in expression is rarely biologically meaningful. The 2-fold threshold (|log2FC| >= 1) is the standard convention in DESeq2 workflows and filters to genes with substantial expression changes.

3. DEG Overview: All 6 Comparisons

Comparison	Total Genes	DEGs	Up	Down	% DEG
radiation_effect	25,519	822	764	58	3.2%
genotype_interaction	26,072	433	76	357	1.7%
GCR40	22,178	2	1	1	0.0%
GCR80	22,178	11	11	0	0.1%
Cs137_100cGy	23,667	229	220	9	1.0%
Cs137_10cGy	23,667	87	79	8	0.4%

Observation: Radiation Type Matters

Co-60 gamma radiation at 100 Gy produces the most DEGs (822), while simulated GCR at 40-80 cGy produces almost none (2-11). This is expected: 100 Gy is 1,000-2,500x higher dose than 40-80 cGy. The Cs-137 comparisons show a clear dose-response (87 at 10 cGy vs 229 at 100 cGy). The sog1-1 interaction DEGs are predominantly downregulated (357/433 = 82%), opposite to the radiation main effect (764/822 = 93% upregulated).

4. Tissue-Specificity: The Tau Index

Since all 6 experiments used whole seedlings, the DEG data has no tissue information. To determine which tissues are responding to radiation, we use the AtGenExpress developmental expression atlas (Schmid et al. 2005), which profiled 237 samples across 33 sub-tissues of Arabidopsis.

The Tau Tissue-Specificity Index

Tau (tau) measures how specifically a gene is expressed in one tissue versus others. It ranges from 0 (expressed equally everywhere) to 1 (expressed in only one tissue). We use the threshold tau >= 0.6 to classify genes as tissue-specific.

```
Formula: tau = sum(1 - x_i / max(x)) / (n - 1)
```

> **For Beginners: What is Tau?**
>
> Imagine a gene that is only turned on in roots. Its expression in roots is high, but in leaves, flowers, and seeds it is near zero. This gene would have a Tau value close to 1 (very tissue-specific). A gene that is equally expressed everywhere (like a housekeeping gene) would have Tau close to 0. We use 0.6 as the cutoff: genes above 0.6 are 'tissue-specific'.

Advanced Note: Tau vs other specificity metrics

Tau is preferred over alternatives (Gini coefficient, entropy-based measures) because it is robust to noise, bounded [0,1], and does not require arbitrary expression thresholds. The 0.6 cutoff is consistent with Yanai et al. (2005) and Kryuchkova-Mostacci & Robinson-Rechavi (2017). In our atlas: 12,772 tissue-specific genes, 8,061 constitutive, across 5 broad organs and 33 sub-tissues.

5. Code Walkthrough: Key Steps

Step 1: Load and threshold DESeq2 results

```
dt[, deg_flag := ifelse(padj < 0.05 & abs(log2FoldChange) >= 1, "yes", "no")]
```

This applies the dual threshold to each of the 6 DESeq2 output tables. Genes passing both criteria are flagged as DEGs.

Step 2: Compute Tau from the atlas

```
tau <- apply(expr_matrix, 1, function(x) sum(1 - x/max(x)) / (n-1))
```

For each gene, Tau is computed across all 33 sub-tissues. Genes with Tau >= 0.6 are classified as tissue-specific and assigned to their predominant tissue (highest expression).

Step 3: Merge DEGs with tissue annotations

```
dt_ann <- merge(deg_table, tissue_ann, by = "gene_id")
```

Each DEG is classified as tissue-specific (assigned to a broad organ + sub-tissue), constitutive (expressed everywhere), or unannotated (not in the atlas, including organellar genes).

Step 4: Statistical tests

Four tests per comparison: (1) chi-square of independence (organ x DEG status), (2) per-tissue Fisher's exact (organ vs rest), (3) pairwise Fisher's exact (10 organ pairs), (4) direction chi-square (organ x up/down). All with Bonferroni correction.

Step 5: Cross-comparison tests

Cochran-Mantel-Haenszel (CMH) test stratified by organ: compares DEG rates between radiation types (Co-60 vs Cs-137) and doses (10 vs 100 cGy). Chi-square tests compare organ distributions and direction patterns across comparisons.

6. Key Results

Per-Comparison Tissue Enrichment

Comparison	Chi-sq	p-value	Enriched	Depleted
radiation_effect	233.6	2.2e-49	Seedling (12.7%)	Root, Flower, Seed
genotype_interaction	14.4	0.006	Flower (3.1%)	Leaf_Shoot
GCR40	SKIP	-	Too few DEGs (2)	-
GCR80	SKIP	-	Too few DEGs (4)	-
Cs137_100cGy	53.9	5.7e-11	Seedling (3.0%), Root (2.4%)	Flower, Leaf_Shoot
Cs137_10cGy	44.4	5.3e-9	Seedling (1.6%), Root (0.9%)	Flower

Cross-Comparison Tests

Test	Statistic	p-value	Interpretation
CMH: Co-60 vs Cs-137	173.7	1.1e-39	Co-60 = 3.1x more DEGs
CMH: Cs-137 10 vs 100cGy	52.0	5.6e-13	100cGy = 2.8x more DEGs
Chi-sq: Genotype organ dist.	87.1	5.5e-18	Different tissue profiles
Chi-sq: Direction (rad vs int.)	456.5	2.8e-101	Up-dominant vs down-dominant
Chi-sq: Overall (4 comparisons)	209.2	4.0e-38	Profiles differ across conditions

Key Finding: sog1-1 Dampens the Radiation Response

The genotype x radiation interaction DEGs are 82% downregulated, while the radiation main effect is 93% upregulated. This means sog1-1 (a DNA damage response mutant) reduces the plant's ability to upregulate radiation-responsive genes. The tissue distribution also differs significantly (chi-sq p=5.5e-18), with Flower and Seed_Silique showing the most interaction DEGs.

7. Methodological Lesson: Old vs New Comparison

For the OSD498_510 radiation_effect comparison, we can directly compare the old (normalized data, no LFC cutoff) and new (unnormalized data, |LFC| >= 1) results to quantify the impact of both corrections.

Metric	Value
Old DEGs (normalized, padj<0.05)	6,942
New DEGs (unnormalized, padj<0.05 & |LFC|>=1)	822
Shared DEGs (in both)	787
Lost DEGs (old=yes, new=no)	6,155
Gained DEGs (old=no, new=yes)	8
Direction flips (among shared)	0
Lost due to LFC threshold (|log2FC_old| < 1)	6,141 (99.8%)
Lost due to normalization change	14
log2FC correlation (shared DEGs)	0.95
baseMean correlation (Spearman)	1.00

What This Tells Us

The 88% reduction in DEGs is almost entirely from the stricter fold-change threshold, not from the normalization correction. Only 14 genes were lost specifically due to switching from normalized to unnormalized counts. The 787 shared DEGs have a log2FC correlation of 0.95 and zero direction flips, confirming that the biological signal is consistent - the corrections made the results more stringent, not different.

8. Output Files Guide

Per-Comparison Outputs

File Pattern	Description
tissue_specific_degs_v2//	Hierarchical CSV folders (5 organs + sub-tissues)
tissue_deg_summary_.csv	Hierarchical summary: organ x sub-tissue DEG counts
tissue_enrichment_stats_.csv	Fisher's exact test results per organ
pairwise_comparisons_.csv	Pairwise Fisher's tests (10 organ pairs)
fig1-5_*_.png	5 figures per comparison (bar, volcano, heatmap, UpSet, forest)

Cross-Comparison Outputs

File	Description
cross_comparison_tissue_summary.csv	DEG counts: organ x comparison matrix
cross_comparison_direction.csv	DEG rates: organ x comparison matrix
cross_comparison_statistics.csv	All Fisher's exact results across 6 comparisons
cross_comparison_statistical_tests.csv	CMH and chi-square cross-comparison tests
fig_cross1-4_*.png	4 cross-comparison figures
old_vs_new_comparison_OSD498_510.csv	Gene-level old vs new comparison

R Script

tissue_specific_deg_analysis_v2.R - Complete parameterized R script covering all 12 sections of the analysis. Can be re-run with different thresholds by changing the configuration variables at the top.

9. How to Interpret the Results

Reading the Tissue Enrichment Forest Plot

Each forest plot (fig5) shows the DEG rate (%) for each organ with 95% confidence intervals. Organs to the right of the dashed line (overall mean) have higher DEG rates. Significance markers: *** ($p<0.001$), ** ($p<0.01$), * ($p<0.05$), ns (not significant), after Bonferroni correction.

Reading the Cross-Comparison Heatmap

The tissue response heatmap (fig_cross1) shows DEG rates for each organ across all 6 comparisons. Darker colors = more DEGs. This reveals which tissues respond most to each radiation condition. Seedling shows the highest rate (11.96%) under Co-60 100Gy, while GCR conditions produce almost no tissue-specific DEGs.

Reading the Dose-Response Plot

The dose-response figure (fig_cross3) plots DEG count vs dose for each organ, separated by radiation source (Cs-137 vs GCR). For Cs-137, all organs show a clear increase from 10 to 100 cGy. For GCR, the increase from 40 to 80 cGy is minimal, reflecting the very low DEG counts at these doses.

> **Caveat: Low-DEG Comparisons**
>
> GCR40 (2 DEGs) and GCR80 (11 DEGs) have insufficient DEGs for meaningful tissue enrichment statistics. Chi-square and direction tests were skipped for these comparisons. This is a real biological signal (low-dose mixed-particle radiation produces minimal 2-fold transcriptional changes), not a data quality issue. Consider lowering the LFC threshold for these comparisons if exploratory analysis is desired.

10. Exercises

Beginner Exercises

1. Open the tissue_deg_summary_radiation_effect.csv file. Which sub-tissue has the most DEGs? Is it up- or down-regulated?

2. Look at fig1 for Cs137_100cGy. Which organ has the most DEGs? Compare this to fig1 for radiation_effect - how does the organ distribution differ between Co-60 and Cs-137 radiation?

3. In the old_vs_new comparison, what percentage of old DEGs were lost because their fold change was below the new threshold?

Intermediate Exercises

4. The CMH test for Co-60 vs Cs-137 gives OR=0.328. Interpret this odds ratio: what does it mean biologically?

5. The genotype_interaction DEGs are 82% downregulated while radiation_effect is 93% upregulated. What does this tell you about the role of SOG1 in the radiation response?

6. Compare the tissue enrichment results for Cs137_10cGy and Cs137_100cGy. Does the enriched tissue change with dose, or just the magnitude?

Advanced Exercises

7. The chi-square test for genotype organ distribution gives X2=87.1, p=5.5e-18. Examine the residuals: which organs contribute most to this significant result?

8. Re-run the analysis with |log2FC| >= 0.5 instead of 1.0. How many more DEGs do you get for GCR40 and GCR80? Does this change the tissue enrichment results?

9. The baseMean correlation between old and new is 1.0 (Spearman), but the log2FC correlation is 0.95. Why might log2FC be less correlated than baseMean when switching from normalized to unnormalized counts?